▯ Easy (10 MCQs)

1. **Which of the following is used to verify file integrity?**
   A) Encoding
   B) Encryption
   C) Hashing
   D) Compression
   Answer: C

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2. **Which hash function produces a 128-bit output?**
   A) SHA-256
   B) SHA-1
   C) MD5
   D) SHA-512
   Answer: C

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3. **A hex editor allows viewing data in:**
   A) English
   B) ASCII only

C) Hexadecimal format
D) Graphical icons
Answer: C

4.

Which process is used to make data unreadable without a key?
A) Hashing
B) Encoding
C) Encryption
D) Compression
Answer: C

5.

Which tool is commonly used as a hex editor?
A) Chrome
B) HxD
C) Notepad
D) Paint
Answer: B

6.

Bit rot is related to:
A) Malware
B) Antivirus
C) Data degradation
D) Encryption speed
Answer: C

7.

Hash collisions occur when:
A) Two inputs give same hash
B) A hash is encrypted
C) File is compressed
D) Two files merge
Answer: A

8.

Which one is reversible?
A) Hashing
B) Encryption
C) Bit rot
D) None

Answer: B

9.

Base64 is an example of:
A) Hashing
B) Encryption
C) Encoding
D) Decryption
Answer: C

10.

Which hash is stronger?
A) MD5
B) SHA-1
C) SHA-256
D) None
Answer: C

Medium (20 MCQs)

11.

A hex value FF in binary is:
A) 00000000
B) 11111111
C) 10101010
D) 10000000
Answer: B

12.

MD5 is considered:
A) Highly secure
B) Vulnerable to collisions
C) Faster than all
D) Unused in forensics
Answer: B

13.

Which concept is NOT related to hashing?
A) Fixed output
B) One-way function

C) Encryption key
D) Collision resistance
Answer: C

14.

Hashing DLs is used to:
A) Change website layout
B) Verify downloads
C) Compress files
D) Encode links
Answer: B

15.

The danger of MD5 collision is:
A) Larger file sizes
B) Data recovery problems
C) Forged identities
D) Faster speed
Answer: C

16.

Bit rot can be minimized by:
A) Running malware
B) Not saving data
C) ECC-enabled systems
D) Formatting weekly
Answer: C

17.

Which is NOT a use of hex editor?
A) Analyzing headers
B) Video rendering
C) File carving
D) Memory analysis
Answer: B

18.

Which statement is true?
A) Hash collisions are harmless
B) All encoding is secure
C) Bit rot affects data slowly
D) Encryption makes file smaller

Answer: C

19.

SHA-256 generates how many bits?
A) 128
B) 160
C) 256
D) 512
Answer: C

20.

Digital evidence integrity is validated using:
A) Antivirus
B) Hash comparison
C) File names
D) Wireshark
Answer: B

21.

The main reason MD5 is no longer preferred:
A) It’s expensive
B) Slow performance
C) Collisions exist
D) Not available freely
Answer: C

22.

The process of converting data to 0x64 0x61 0x74 0x61 is called:
A) Hashing
B) Base64 encoding
C) Hexadecimal conversion
D) Compression
Answer: C

23.

Which algorithm is most collision-resistant?
A) MD5
B) SHA-1
C) SHA-256
D) DES
Answer: C

24.

A byte in hex has how many characters?
A) 1
B) 2
C) 3
D) 4
Answer: B

25.

Bit rot affects which type of files?
A) Only images
B) Only text
C) Any file
D) Only executables
Answer: C

26.

Which is a result of hash collision?
A) Faster processing
B) File duplication

C) Legal evidence compromise
D) Increased file size
Answer: C

27.

Which is best for password hashing?
A) SHA-1
B) bcrypt
C) MD5
D) DES
Answer: B

28.

Hex editors help analyze:
A) Source code only
B) Registry values
C) Low-level file structure
D) Browser history
Answer: C

29.

In forensic tools, hashing helps with:
A) Drawing data
B) Verifying evidence integrity
C) Formatting logs
D) Generating IPs
Answer: B

30.

When files degrade over time, it is called:
A) File loss
B) Bit rot
C) CRC error
D) Compression failure
Answer: B

Hard (10 MCQs)

31.

Which of the following is MOST likely to cause a hash mismatch?
A) File renaming

B) File moving
C) Data tampering
D) View-only access
Answer: C

32.

Hex editor analysis is crucial when:
A) File is too large
B) Extension is missing
C) File is encrypted
D) File is hashed
Answer: B

33.

What is the output size of SHA-512 in hex characters?
A) 128
B) 64
C) 256
D) 512
Answer: A *(SHA-512 → 512 bits = 128 hex chars)*

34.

How do hash collisions challenge forensic evidence?
A) Cause virus detection
B) Corrupt files
C) Make evidence unreliable
D) Stop hashing
Answer: C

35.

What is the risk in relying on MD5 for verification?
A) Long computation
B) False duplicates
C) Untrusted vendors
D) Misleading file names
Answer: B

36.

Which of the following is most secure?
A) MD5
B) SHA-1
C) SHA-3
D) SHA-0

Answer: C

37.

If you get the same hash for two files, you must:
A) Ignore it
B) Accept both
C) Check for collision
D) Delete one
Answer: C

38.

Which condition is NOT linked to bit rot?
A) Age of disk
B) Magnetism
C) CRC mismatch
D) AES encryption
Answer: D

39.

In legal forensics, a weak hash algorithm can:
A) Increase trust
B) Be accepted blindly
C) Be challenged in court
D) Speed up disk imaging
Answer: C

40.

Which forensic task benefits MOST from hashing?
A) Chain of custody
B) Malware execution
C) Evidence validation
D) File renaming
Answer: C